

Kartik Kurhewar

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📍 Pune, India

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with **4+ years of experience** shipping **production GenAI healthcare products** end-to-end — **RAG pipelines** and **LLM integration** on **Python/FastAPI** backends, paired with **React.js** frontends. Experienced across **retrieval, chunking, hybrid search, reranking, and multimodal RAG**, plus **REST API design, authentication & security (JWT, RBAC)**, and **performance optimization**, deploying to **Google Cloud Platform (GCP)** and **Microsoft Azure** via automated **CI/CD pipelines**.

EXPERIENCE

- Cognizant** Pune, India
Software Engineer Dec 2021 – Present
 - Full-Stack GenAI Product Delivery:** Built and maintained full-stack features for **production GenAI healthcare products**, connecting **React.js** frontends to **Python/FastAPI** backends; used **GitHub Copilot** and **Claude Code** to accelerate development velocity.
 - RAG Pipelines & Retrieval:** Built **RAG (Retrieval-Augmented Generation)** pipelines over domain-specific healthcare data using **LangChain** and **ChromaDB** — **recursive and semantic chunking** with tuned overlap, **embeddings**, and **cosine-similarity retrieval** — with **prompt engineering** and context grounding to keep **LLM** responses accurate.
 - Advanced Retrieval & Reranking:** Improved retrieval quality with **hybrid search (dense embeddings + BM25)**, **cross-encoder reranking**, **multi-query retrieval**, and **RAG-Fusion (Reciprocal Rank Fusion)**; extended coverage to tables and images with **multimodal RAG (Unstructured.io)**.
 - Backend & Database Development:** Designed and built **RESTful APIs** and backend services using **Python** and **FastAPI**, modeling **PostgreSQL** schemas and optimizing queries to support frontend data requirements and internal tooling.
 - Frontend Development & Performance:** Developed **pixel-perfect, responsive user interfaces** from Figma designs using **React.js**, **HTML5**, **CSS3**, **Bootstrap**, and **Tailwind CSS**, building reusable components with **Hooks**, **Context API**, and **Redux Toolkit** (cutting the design-to-development cycle by **25%**); improved performance by **30%** via **lazy loading**, **code splitting**, memoization, and bundle optimization across Chrome, Firefox, and Safari.
 - Authentication & Security:** Implemented secure authentication and authorization using **JWT** and **Auth0** across frontend and backend, enforcing **role-based access control (RBAC)** for secure data access.
 - Cloud Deployment & CI/CD:** Deployed React applications on **Google Cloud Run** and **Microsoft Azure** through automated **GitHub Actions CI/CD pipelines**, reducing deployment time by **40%**.
 - Testing & Code Quality:** Refactored legacy React codebases, implemented **unit testing**, and enforced coding standards, reducing user-reported issues by **35%**.
 - Collaboration & Agile Development:** Mentored **3 junior developers** and collaborated with cross-functional Agile teams, reducing onboarding time by **50%** while improving sprint delivery.

TECHNICAL SKILLS

- Languages:** JavaScript, Python, SQL, HTML5, CSS3
- Generative AI:** RAG, RAG-Fusion, Hybrid Search (Dense + BM25), Reranking (Cross-Encoders), Multi-Query Retrieval, Semantic Chunking, Vector Search, Embeddings, LangChain, ChromaDB, Multimodal RAG (Unstructured.io), Prompt Engineering
- Backend & Databases:** FastAPI, Pydantic, REST APIs, PostgreSQL
- Frontend Frameworks & Libraries:** React.js, Redux Toolkit, Context API, React Router, Tailwind CSS, Bootstrap, Axios, Vite, Webpack
- Architecture & Design:** Reusable Component Design, State Management, Performance Optimization, Micro-Frontends
- Authentication & Security:** JWT, Auth0, Role-Based Access Control (RBAC)
- DevOps & Cloud:** Git, GitHub, GitHub Actions, CI/CD, Docker, Google Cloud Platform (GCP), Microsoft Azure
- Testing:** Unit Testing, Usability Testing

EDUCATION

- Priyadarshini College of Engineering** Nagpur, India
Bachelor of Engineering in Information Technology July 2017 – Aug. 2021